Programming basics and origin

23 May 2026 11:45

~~OOPS~~

Arduino

Systems language

C

1972 · Dennis Ritchie, Bell Labs

1972

WHY IT WAS CREATED

To rewrite the Unix operating system in a portable way — previously OS code was written in assembly, which was hardware-specific and painful to move.

WHY IT'S USED TODAY

It's the closest you can get to hardware without writing assembly. Extremely fast and lightweight — used wherever performance is non-negotiable.

USED IN

OS kernels · Embedded systems · Microcontrollers · Compilers

compilers. OS

Systems + OOP

C++

1983 · Bjarne Stroustrup, Bell Labs

1983

WHY IT WAS CREATED

C was great but lacked structure for large programs. Stroustrup added object-oriented programming (classes, inheritance) on top of C — literally called "C with Classes" at first.

WHY IT'S USED TODAY

Gives C's raw speed with modern programming structures. The go-to language when you need both high performance and organized code.

USED IN

Game engines · Competitive programming · Browsers · Finance systems

Unity → Dr Driving

General purpose

Python

AI/ML C C++

1991 · Guido van Rossum, Netherlands

pandas, th

WHY IT WAS CREATED

Guido wanted a language that was easy to read and write — almost like plain English. He prioritized developer happiness over raw speed, which was a radical idea at the time.

WHY IT'S USED TODAY

Minimal syntax means you write less code to do more. Dominates AI/ML because of its massive library ecosystem (NumPy, TensorFlow, PyTorch).

USED IN

AI / ML · Data science · Web backends · Scripting

FASTAPI

Platform-independent

Java

1995 OOPS

1995 · James Gosling, Sun Microsystems

JVM

WHY IT WAS CREATED

The internet was exploding in the 90s and code written for one OS wouldn't run on another. Java's motto: "Write once, run anywhere" — it runs on a virtual machine (JVM) instead of directly on hardware.

WHY IT'S USED TODAY

Extremely robust for large enterprise systems. Android apps are built in Java (or Kotlin, its modern cousin). Strong type system catches bugs early.

USED IN

Android apps · Enterprise software · Banking systems · Big data

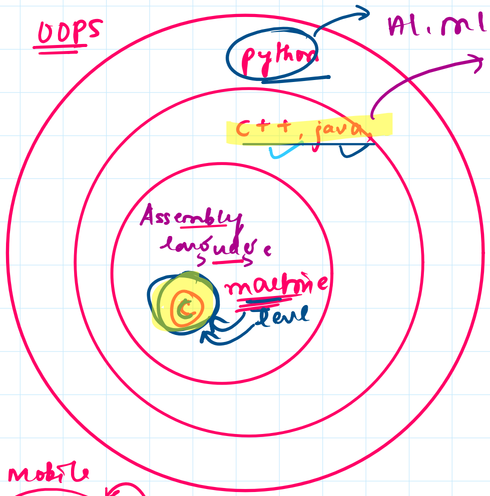
How they compare

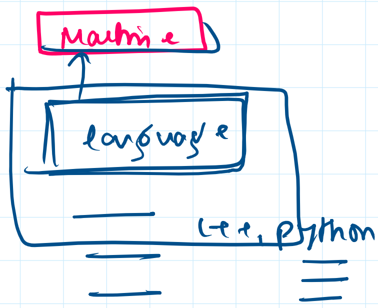
game dev

FEATURE	C	C++	PYTHON	JAVA
Speed	Fastest	Very fast	Slowest	Fast (JVM)
Ease of learning	Hard	Hard	Easiest	Moderate
Memory control	Manual	Manual	Automatic	Automatic
OOP support	None	Full	Full	Full
Type system	Static	Static	Dynamic	Static
Best for juniors	After C++	DSA & CP	First language	App dev

languages

CP

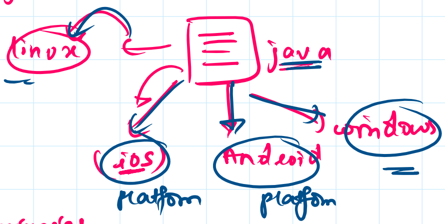




mobile

kotlin → google → Android

java



java, C++, C → #include <stdio.h>

python → import pandas as py

tensorflow

AI

Variables, data types

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- Int, → integer (+1, 0, -1)
- Float, → float (decimal) = 9.09, 10.01 .. etc
- String, → group of characters = "Hello world" → character
- Char,
- Byte,
- Unsigned int,
- Long int,

int | data

x → container = variables

datatypes

char ch = "H"

variable

character

datatype

float m

4 byte float x = 9.09

double → 8 bytes

Byte a = -127 to 128

unsigned (-/+)

unsigned int x = +178

long int = +2147 8958178

limit ↑↑ + +

→ long int = 214789 ...11

int x = (10)

char ch = "H"

float, string, char, long → int

10 → data

x, y, z, container = variables

datatypes

Conditional Statements if, else

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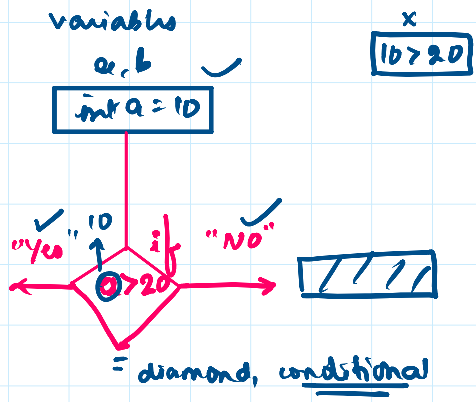
If,
If/else,
Else if,
Switch,
Ternary operator

int a = 10;

datatype — variable — value



C → int a = 10;
compiler → if (a > 20){
10 < cout << "A is great";
}
Else {
cout << "A is less";
}

10
a
10>20
"A is less"

Example 1 : basic CGPA

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10 = O

C , C++ , python , java

o/p → printf   cout   print   system.out.println

i/p → scanf   cin   input()   Scanner sc   sc.nextLine()

6.5
6
7.5
8.5

if / else ladder

```
float user = 9.5
if(user > 9){
    cout<< "Outstanding = O"<<endl;
}
else if( user < 9 && user >= 8)
         8.5             8.5   false
else if( true && false ){
          1 && 1 = 1 = ~~true~~ false

    cout<< "A"
else if ( user < 8 && user > 7){

}
```

6

>= , <= } Relational operations

= , == } Assignment

Arithmetic { + , - , % → **

Example 2: Score check

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Game → $\boxed{\text{PUBG}}$

10 players, Each player should shoot 100 enemies

Add on "score"++;

```
int players = 10;        → [1] player    → if(player >= 100){        → if/else.
int enemies = 100;                           score ++;
              200                            score --;
int score = 0;
```

0, 1, 2, 3, 4

```
int score = 0;
score++;
cout<<score;
```

int → score, string → player

score	player
—	"A"
—	"B"
—	"C"
—	"D"
	..

mp;

player → enemies → score += 2

→ fool → score -= 2

maps, Hashmaps { Datastructure

```
unordered_map<int, string> mp;
int enemies = 100;
int fool;
for( auto i : player){
        if(enemies >= 100){
              score += 1;
        }
}
```

loops

Questions

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Q.1

```cpp
// Online C++ compiler to run C++ program online
#include <iostream>
using namespace std;
int main() {
    int x = 5;
    if(x == 5)
        cout<<"X is equal to 5"<<endl;

    return 0;
}
```

5

x

== ; =

Equality assignment

$x == 5$ int $x = 10$

int $x = 10$

if $(x == 5)\{$

10

}

Else{

}

Q.2

```cpp
// Online C++ compiler to run C++ program online
#include <iostream>
using namespace std;
int main() {
    int x = 5;
    if(x == 4);
        cout<<"X is equal to 5"<<endl;

    return 0;
}
```

Aptitude,

O/p: 0, 'not generate any o/p';

Q.3.

```cpp
// Online C++ compiler to run C++ program online
#include <iostream>
using namespace std;
int main() {
    int x = 3;
    if(x == 5)
        cout<<"X is equal to 5"<<endl;
    else
        cout<<"X is not equal to 5"<<endl;
    return 0;
}
```

A1: x is not equal to 5.

Q.4.

main.cpp

```cpp
// Online C++ compiler to run C++ program online
#include <iostream>
using namespace std;
int main() {
    float x = 0.5;
    if(x==0.5)
        cout<<"abc"<<endl;
    else
        cout<<"def"<<endl;
    return 0;
}
```

float x = 0.5;

abc

Array basics

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What is array, initialization, printing array and so on.,

Array is a data structure
basic

→ * collection of data elements which each elements are of data type
→ initializate [ ]
→ simple way to store many elements of same data type.

* int [1, 2, "Array"];

↗ initializae C, C++, SEE,,

int arr[10] = {1, 2, 3, 4, ... 10, 11};

size

datatype

variable

data elements

op ; ??

java → * Array bound of Exeption

→

variable pointing on Array :

1000	1004	1008	1012	1016	1020	1024
1	2	3	4	5	6	7

int arr[7]; =

indexing → 0 1 2 3 4 5 6

int = 4 bytes

4×6 = 24

{1, 2, 3, 4, 5, 6, 7} = 24 bytes

1 byte = 7bit

arr[0] = 1
arr[1] = 2
arr[2] = 3
arr[3] = 4

variable pointing on Array :

1000	1004	1008	1012	1016	1020	1024
1	2	3	4	5	6	7

int arr[7]; =

indexing → 0 1 2 3 4 5 6

" loop "

→ multiple times
→ until the condition
terminal
satisfied

loop

for loop | while loop | do-while loop.

for(int i = 0; i < n; i++){ → incrementation.
score ++,

if(<)

loop

for(int i = 0; i<n; i++){   → incrementation
score ++;
--
}

initial

condition

if(< >)

**

loop: for (initialization; condition; incrementation)   → generalization

while

int i = 10;

while(condition)
i++;
i++; ____
i++ ____
};

flexiblity

while (i<20){
i++;
____
____
};

do {
____
____
____ incrementation (i++);
}while (condition)

int i = 5;

do{
1st iteration
____ loop runs for
____ 1st iteration
____
}while(~~i~~ 5 >10);

```
int arr[3]= {1, 2, 3}
for(int i=0; i<2; i++){
    cout<<arr[i];
};
```

i = 0 → 1
i = 1 → 2
i = 2 → 3

1	2	3
0	1	2

Loops

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For loops,
While loops,
Do-while loops.

Q.1

```cpp
// Online C++ compiler to run C++ program online
#include <iostream>
using namespace std;
int main() {
    int arr[5] = {1,2,3,4,5};
    cout<<arr[0];

    return 0;
}
```

op: 1

Q.2

```cpp
#include <iostream>
using namespace std;
int main() {
    int arr[5] = {1, 2, 3, 4, 5};
    for (int i = 0; i < 5; i++) {
        cout << arr[i] << " ";
    }
    return 0;
}
```

0
1
2
3
4

1 2 3 4 5

1	2	3	4	5
0	1	2	3	4

1_2_3_4_5

op - 1,2

Q.3

indexing

```cpp
#include <iostream>
using namespace std;
int main() {
    int arr[5] = {1, 2, 3, 4, 5};
    for (int i = 4; i >= 0; i--) {
        cout << arr[i] << " ";
    }
    return 0;
}
```

(a) 5, 4, 3, 2, 1

B) 5, 4, 3, 2

* reverse

1	2	3	4	5
0	1	2	3	4

>=

int i=4

Q.5

```cpp
#include <iostream>
using namespace std;
int main() {
    int arr[5] = {10, 20};
    for (int i = 0; i < 5; i++) {
        cout << arr[i] << " ";
    }
    return 0;
}
```

| 10 | 20 | – | – | – |

10, 20, x00, x00, —

Q.6.

```cpp
#include <iostream>
using namespace std;
int main() {
    int arr[5] = {1, 2, 3, 4, 5};
    int sum = 0;
    for (int i = 0; i < 5; i++) {
        sum += arr[i];
    }
    cout << sum;
    return 0;
}
```

15

0 + 1 + 2 + 3 + 4

sum = 15

15

1	2	3	4	5
0	1	2	3	4

~~10~~ ~~6~~ ~~3~~ x $\boxed{\cancel{0}}$ sum

sum += arr[i]

0
1
2
3
4

Q.7.

```cpp
#include <iostream>
using namespace std;
int main() {
    int arr[5] = {1, 2, 3, 4, 5};
    for (int i = 0; i < 5; i++) {
        arr[i] = arr[i] * 2;
    }
    for (int i = 0; i < 5; i++) {
        cout << arr[i] << " ";
    }
    return 0;
}
```

n

→ time complexity

space complexity = 20 ≅ → O(20) → O(n)

5 × 4 = 20

size

int arr[n]

5

O(5) ≅ O(n)

3, 4, 5
2

for (int i = 1; i <= 5; i++){

for (int i = 1; i <= 5; i++){

O(n^2)

```
for(int i=1; i<=5; i++){
    for(int j=1; j<=5; j++){
```

$5 \times j$

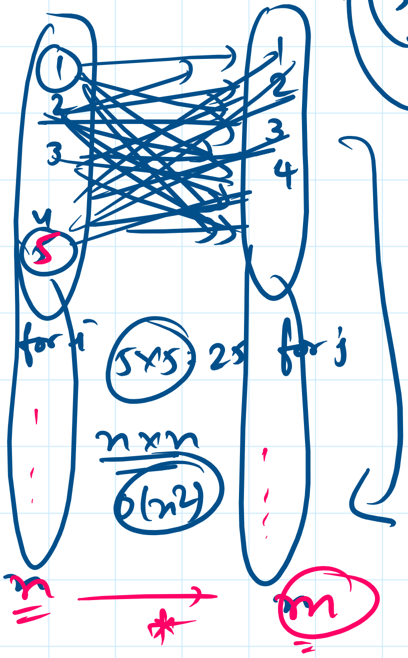
$5 \times j$

$5 \times j$

5

5

}



$5+5+5+5+5 = 25$

$\rightarrow \boxed{O(n*m)} \rightarrow \boxed{O(n*m)}$

$O(n*n) = O(n^2)$